# Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
**Microelectronic Circuits** 8th Edition, *Oxford University Press*, 2019.

- Chapter 10.1 High-Frequency Transistor Model
- Chapter 10.2 High-Frequency Response of CS and CE Amplifier

# Recap

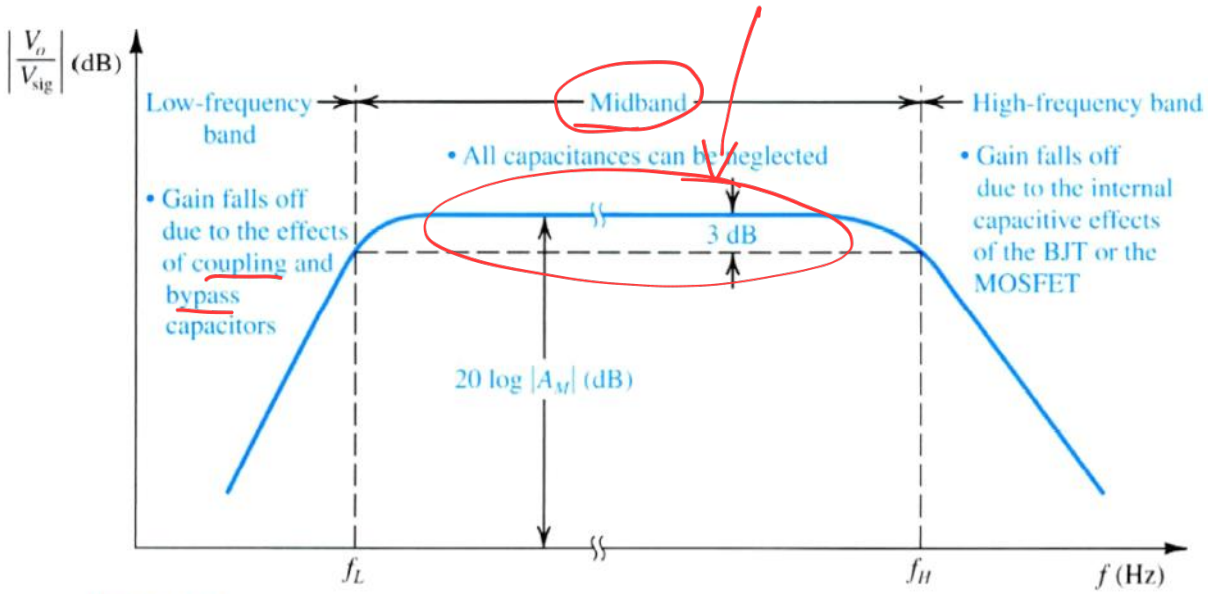



**Figure 10.1** Graph of the gain of a discrete-circuit BJT or MOS amplifier versus the frequency of the input signal, showing the three frequency bands relevant to frequency-response determination.

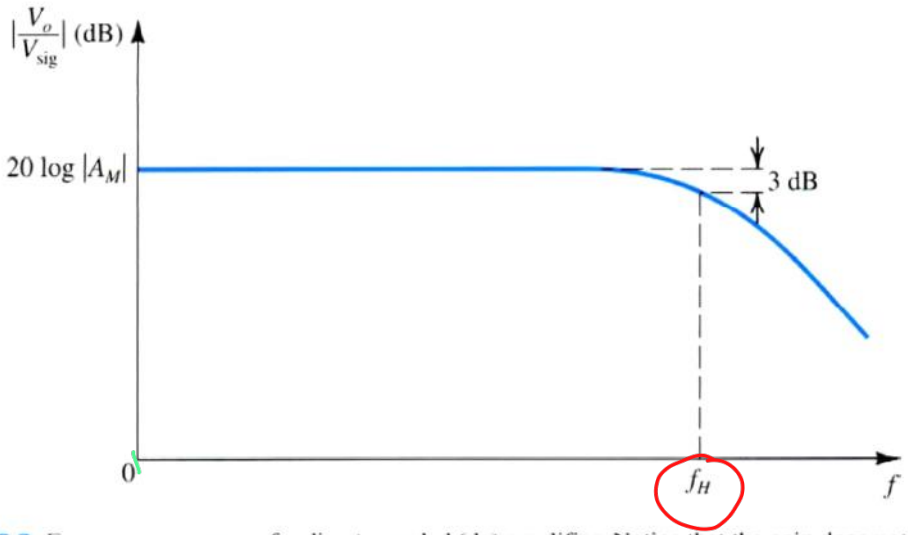
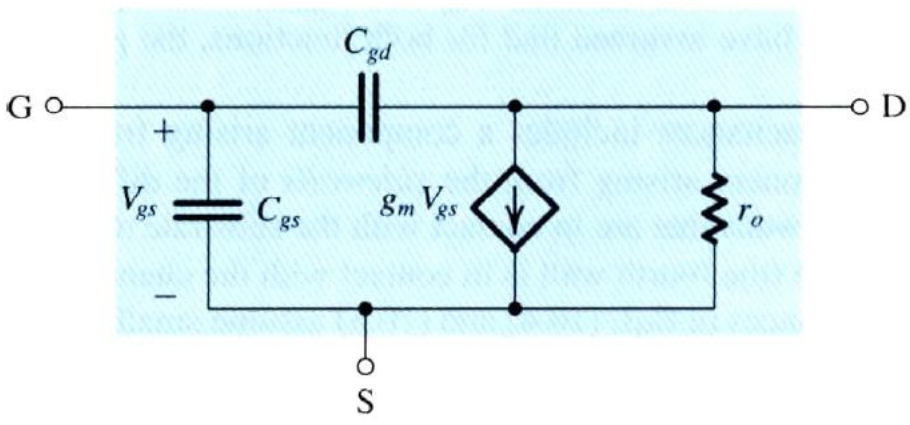
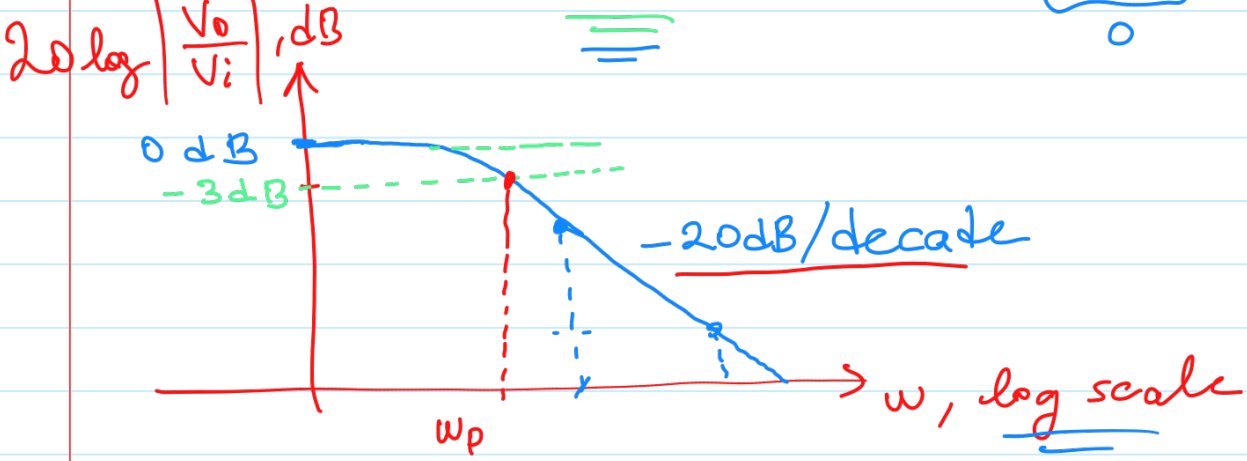


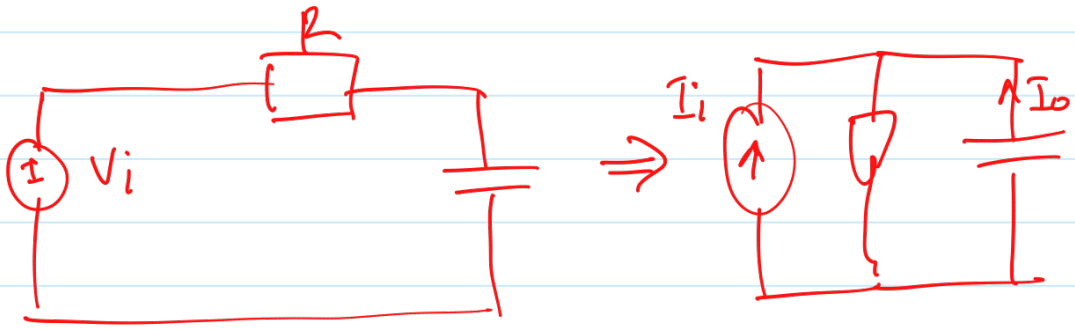

**Figure 10.2** Frequency response of a direct-coupled (dc) amplifier. Notice that the gain does *not* fall off at low frequencies, and the midband gain $A_M$ extends down to zero frequency.



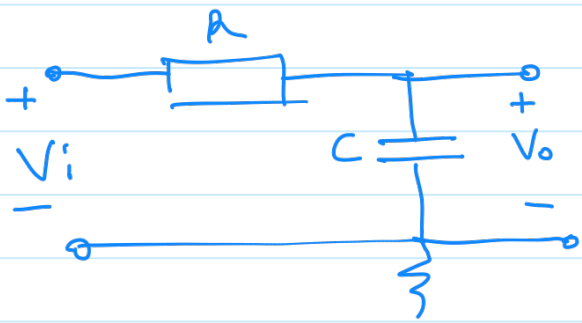
$$\frac{|V_o|}{|V_i|} = \frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_p}\right)^2}} = \frac{1}{1}$$

$$20\log 1 = 0$$



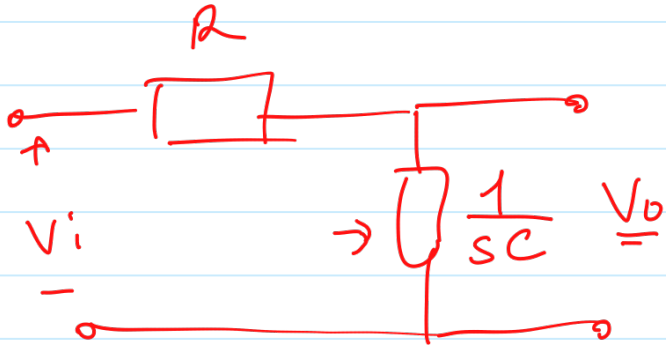



# Frequency Resp. of Low-Pass Single-Time-Constant Circuits




$$\tau = RC$$

$$\omega_p = \frac{1}{\tau} = \frac{1}{RC}$$




$$\frac{V_o}{V_i} = \frac{\frac{1}{sC}}{\left(R+\frac{1}{sC}\right)} \cdot sC$$

$$= \frac{1}{1+sRC}$$

$$\left|\frac{V_o}{V_i}\right| = \left|\frac{1}{1+\frac{j\omega}{\omega_p}}\right| = \frac{|1|}{\sqrt{1^2+\left(\frac{\omega}{\omega_p}\right)^2}} = \frac{1}{1+\frac{s}{\frac{1}{RC}\omega_p}}$$

$$A(\omega) = \frac{|V_o|}{|V_i|} = \frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_p}\right)^2}} \Leftarrow$$

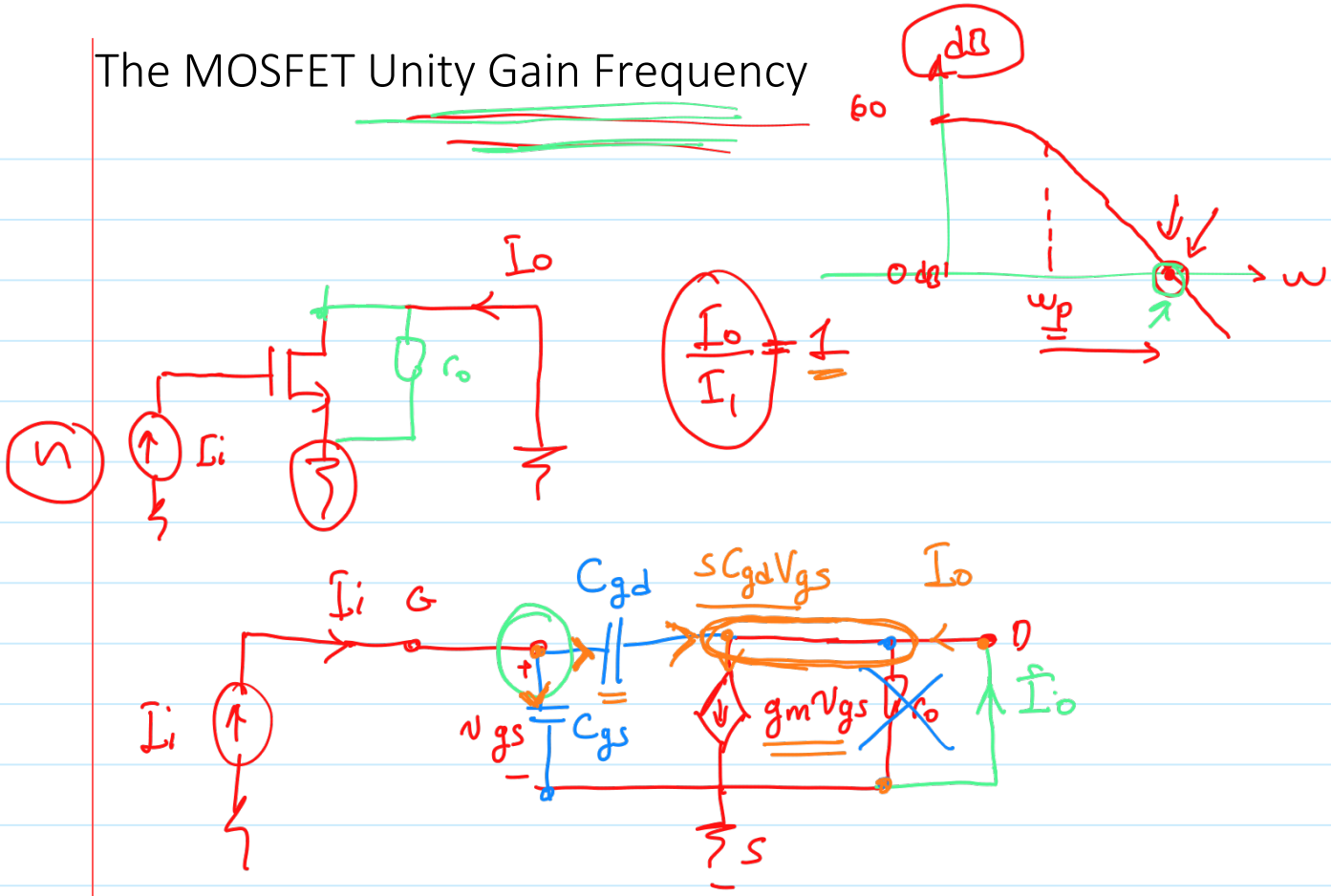
$$A(\omega_p) = \frac{1}{\sqrt{1+1}} = \frac{1}{\sqrt{2}}$$

$$\frac{|V_o|}{|V_i|} = \frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_p}\right)^2}} \quad @ \text{ high } \omega \quad \frac{\omega_p}{\omega}$$

$$20\log\frac{1}{10} = -20\,dB$$

$$\log\frac{1}{10} = -1$$

# The MOSFET Unity Gain Frequency







$$I_o = g_m V_{gs} - sC_{gd}V_{gs} = (g_m - sC_{gd})V_{gs}$$

$$I_i = sC_{gd}.V_{gs} + sC_{gs}.V_{gs} \Rightarrow V_{gs} = \frac{I_i}{s(C_{gd}+C_{gs})}$$

$$\frac{I_o}{I_i} = \frac{(g_m - sC_{gd})}{s(C_{gd}+C_{gs})}$$

$$\frac{I_o}{I_i} = \frac{g_m - j\omega C_{gd}}{j\omega(C_{gd}+C_{gs})} \approx \frac{g_m}{j\omega(C_{gd}+C_{gs})}$$

$$\frac{g_m}{\omega_T(C_{gd}+C_{gs})} = 1 \Rightarrow \omega_T = \frac{g_m}{C_{gd}+C_{gs}}$$

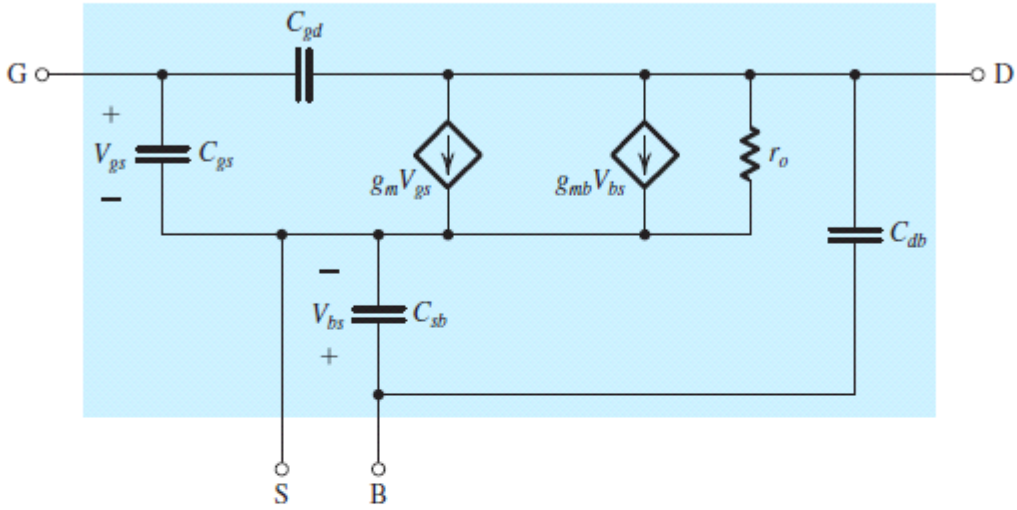
$$\omega = 2\pi f$$

$$f_T = \frac{g_m}{2\pi(C_{gd}+C_{gs})}$$

# Summary Table

**Table 10.1** The MOSFET High-Frequency Model

**Model**




**Model Parameters**

$$g_m = \mu_n C_{ox} \frac{W}{L} |V_{OV}| = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} = \frac{2I_D}{|V_{OV}|}$$

$$C_{sb} = \frac{C_{sb0}}{\sqrt{1 + \frac{|V_{SB}|}{V_0}}}$$

$$g_{mb} = \chi g_m, \quad \chi = 0.1 \text{ to } 0.2$$

$$r_o = |V_A| / I_D$$

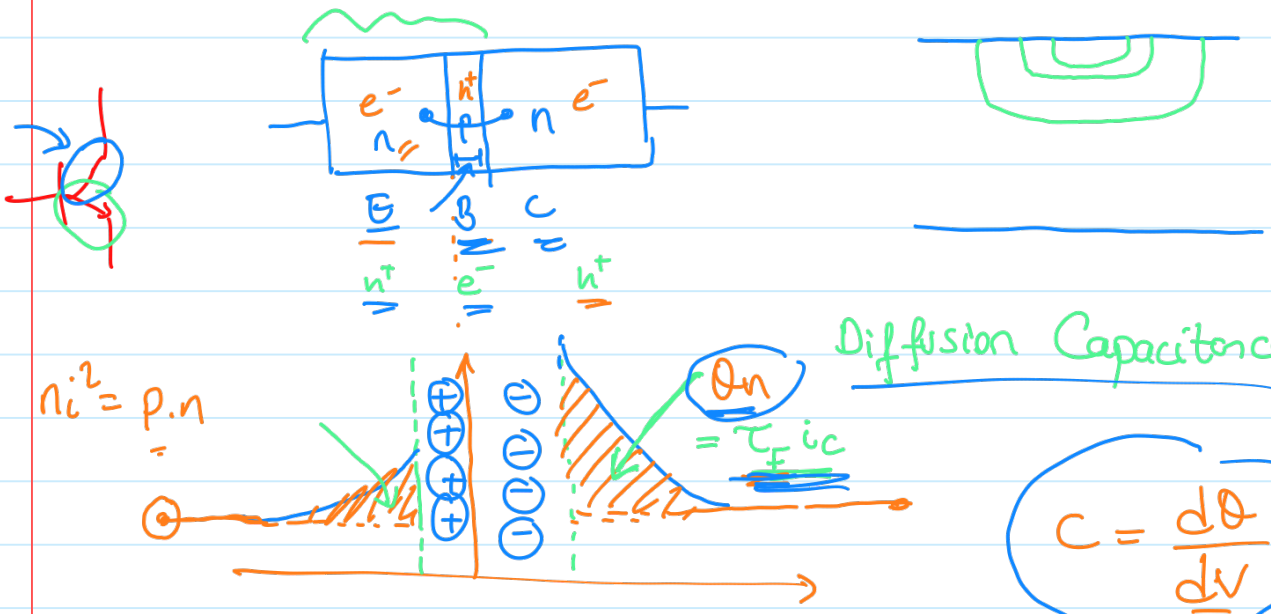
$$C_{db} = \frac{C_{db0}}{\sqrt{1 + \frac{|V_{DB}|}{V_0}}}$$

$$C_{gs} = \frac{2}{3} WLC_{ox} + WL_{ov} C_{ox}$$

$$C_{gd} = WL_{ov} C_{ox}$$

$$f_T = \frac{g_m}{2\pi (C_{gs} + C_{gd})}$$

# The BJT High Frequency Model




$$C = \frac{dQ}{dV}$$

$$C = \tau_F \frac{di_C}{dv_{BE}} = g_m \tau_F = \frac{I_C}{V_T} \tau_F \quad (1)$$

$$C = \frac{d(\tau_F . i_C)}{dv_{BE}}$$

$$C_{je} \approx 2\, C_{je0} \quad (2)$$

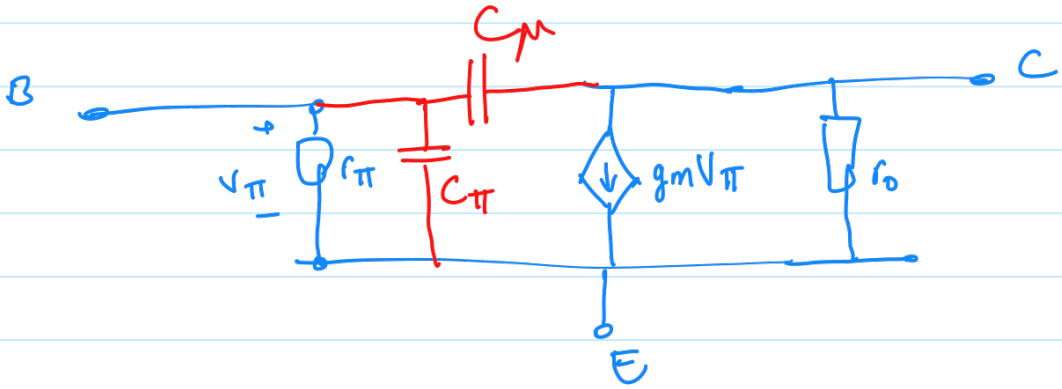
Collector Base Junction Capacitance $C_\mu$

$$C_\mu = \frac{C_{\mu 0}}{\left(1 + \frac{V_{CB}}{V_{0C}}\right)^m} \qquad m \approx 1/2$$

# The BJT High Frequency Model

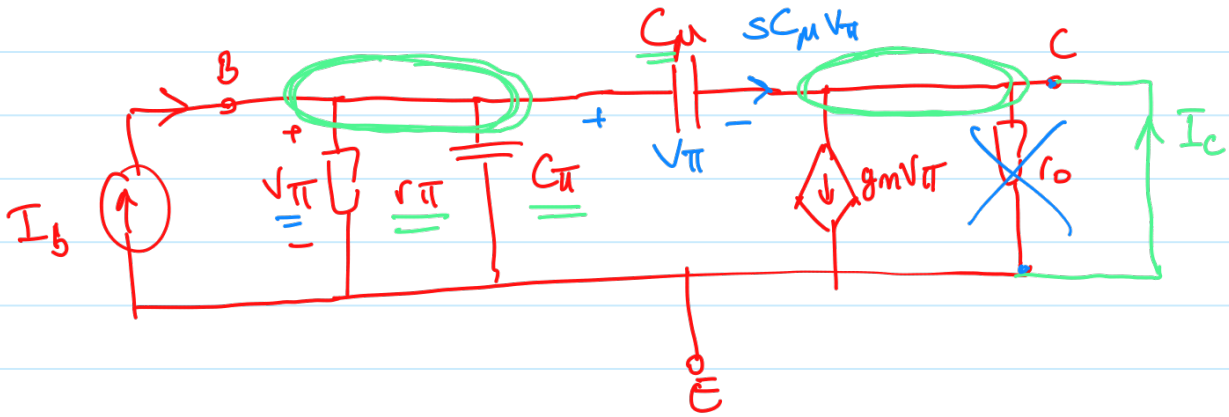
BE → $C_{de}$ + $C_{je}$ = $C_\pi$

BC → $C_\mu$



# The BJT Unity Gain Frequency

$0.2 f_T$ ✓




$$I_c = g_m V_\pi - sC_\mu V_\pi = (g_m - sC_\mu) V_\pi$$

$$I_b = \frac{V_\pi}{\left(\frac{1}{r_\pi} + sC_\pi + sC_\mu\right)^{-1}}$$

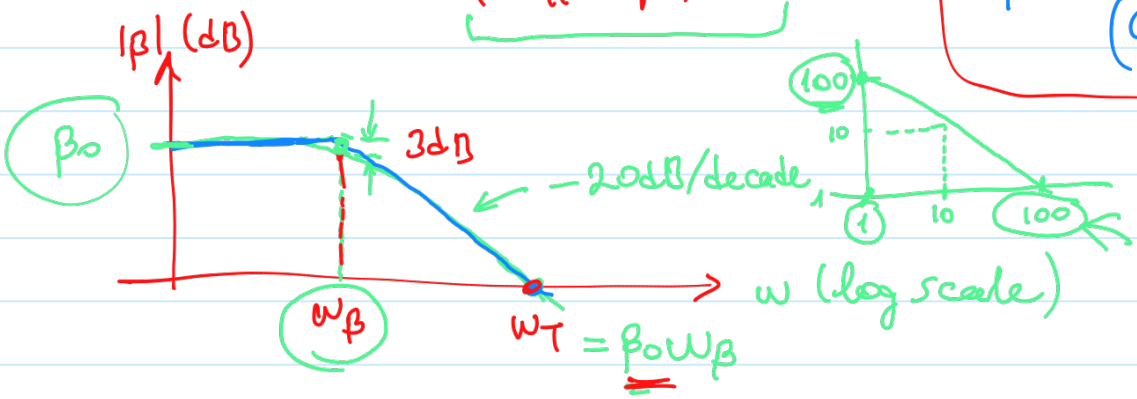
$$V_\pi = \frac{I_b}{\frac{1}{r_\pi} + sC_\pi + sC_\mu}$$

$$\beta = \frac{I_c}{I_b} = \frac{g_m - sC_\mu}{\frac{1}{r_\pi} + sC_\pi + sC_\mu} \approx \frac{g_m}{\frac{1}{r_\pi} + s(C_\pi + C_\mu)}$$

$$\beta = \frac{g_m \cdot r_\pi}{1 + s(C_\pi + C_\mu) r_\pi} = \frac{\beta_o}{1 + s(C_\pi + C_\mu) r_\pi}$$

$\beta_o = \frac{I_C}{V_T} \cdot \frac{V_T}{I_B}$

$$\omega_\beta = \frac{1}{(C_\pi + C_\mu) r_\pi}$$



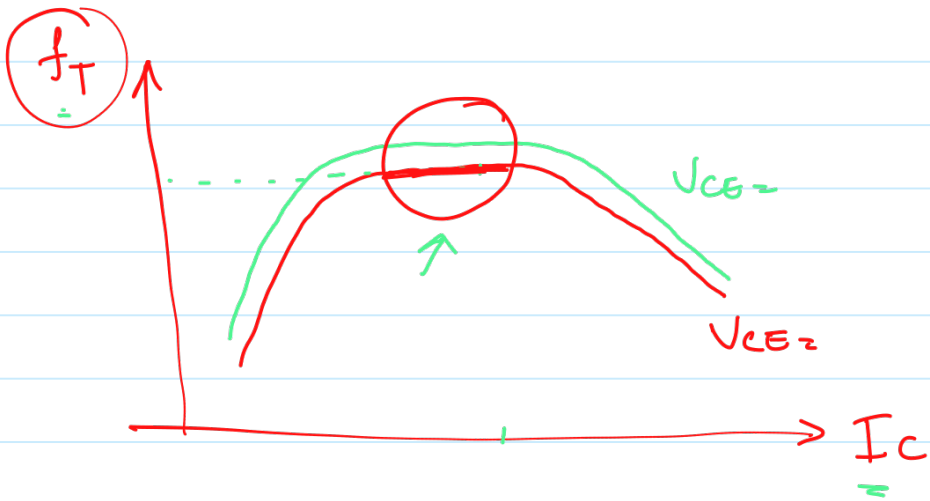
# The BJT Unity Gain Frequency

$$\omega_T = \beta_0 \cdot \omega_\beta \leftarrow \omega_\beta = \frac{1}{(C_\pi + C_\mu) r_\pi}$$

$$= g_m r_\pi \cdot \frac{1}{(C_\pi + C_\mu) r_\pi}$$

$$\omega_T = \frac{g_m}{C_\pi + C_\mu}$$

$$f_T = \frac{g_m}{2\pi (C_\pi + C_\mu)}$$

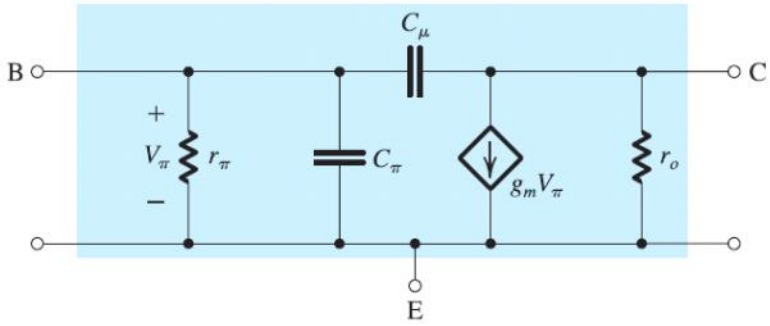



$$f_T = \frac{g_m}{2\pi \cdot (C_\pi + C_\mu)} = \frac{g_m}{2\pi g_m r_E} = \frac{1}{2\pi r_E}$$

$C_\pi \rightarrow C_{de} + C_{je}$

# Summary Table

**Table 10.2** The BJT High-Frequency Model




$g_m = I_C/V_T$ $\quad r_o = |V_A|/I_C$ $\quad r_\pi = \beta_0/g_m$

$C_\pi + C_\mu = \dfrac{g_m}{2\pi f_T}$ $\quad C_\pi = C_{de} + C_{je}$ $\quad C_{de} = \tau_F g_m$ $\quad C_{je} \simeq 2C_{je0}$

$C_\mu = C_{jc0} \Big/ \left(1 + \dfrac{|V_{CB}|}{V_{0c}}\right)^m$ $\quad m = 0.3\text{–}0.5$

# The Common Source Amplifier

Template

Template